

## Supplemental Materials 1 Rare-events logit regression for robustness check

Standard logit regression can underestimate the probability of rare events and may produce finite-sample bias when events are sparse. King and Zeng (2001) developed rare-events logit regression to address this problem by correcting small-sample bias and, when the population event rate is known or assumed, applying either prior or weighting correction. Because model specification may be uncertain in rare-event settings and the sample is sufficiently large (Xie and Manski 1989), this study adopts the weighting correction approach as a robustness check for the substantive findings.

The weighting correction model assigns different weights to event and non-event observations and obtains the maximum likelihood estimates of the parameters by maximizing the following weighted log-likelihood function:

$$\begin{aligned} \ln L_w(\beta|y) &= w_1 \sum_{\{Y_i=1\}} \ln(\pi_i) + w_0 \sum_{\{Y_i=0\}} \ln(1-\pi_i) \\ &= - \sum_{i=1}^n w_i \ln(1 + e^{(1-2Y_i)x_i\beta}) \end{aligned}$$

where  $\tau$  denotes the known population event probability (typically obtained from census or large-sample studies),  $\bar{y}$  denotes the sample event probability,  $w_1 = \tau / \bar{y}$  is the weight assigned to observations with  $Y_i = 1$  (i.e., participation in emergency volunteering), and  $w_0 = (1 - \tau) / (1 - \bar{y})$  is the weight assigned to observations with  $Y_i = 0$  (i.e., non-participation), such that  $w_i = w_1 Y_i + w_0(1 - Y_i)$ . By compensating for sampling bias through these weights, the model aligns the sample distribution with the population distribution and replaces the unweighted log-likelihood function of the standard logit model, thereby adjusting the influence of each observation  $w_i$  on the parameter  $\beta$ . This procedure corrects the bias in event probability estimates that arises when events are rare relative to non-events in standard logit estimation.

Implementing the weighting correction requires specifying the population event probability  $\tau$ . Because  $\tau$  cannot be observed directly in this study, plausible values are

specified using two other sources of information. The first source is the sample event rate ( $y = 5.35\%$ ), which serves as a natural benchmark given the nationally based survey design for CSS 2021 data. To probe robustness, Regolith models are estimated under four values of  $\tau$  derived from this benchmark:  $0.5y$ ,  $y$ ,  $2y$ , and  $3y$ .

The second source consists of two externally informed reference values drawn from official materials. The first,  $12.28\%$ , is reported in the *Report on Development of Voluntary Service in China (2021–2022)*, which, based on survey data, indicates the share of respondents who participated in emergency rescue and disaster relief activities in 2020 (Zhang and Tian 2022). The second,  $1.34\%$ , is an approximate reference value constructed from multiple official sources. *Annual Report on China's Philanthropy Development (2021)* reports a total of 231.13 million registered volunteers in 2020 (Yang and Zhu 2021). Because this figure spans all service domains, the proportion of emergency volunteer teams among all volunteer teams reported on the official China Volunteer Service website (52,299 out of 637,478, and approximately  $8.20\%$ ) is used to approximate the share of emergency volunteers (China Volunteer Service 2021), yielding an estimated 18.96 million. Dividing this figure by China's official 2021 population of 1.4121 billion in *China Statistical Yearbook 2021* (National Bureau of Statistics of China 2021), produces an estimated rate of approximately  $1.34\%$ . As these official materials do not provide a precise population estimate of individuals who actually participated in disaster volunteering, the corresponding values are employed only as supplementary reference points rather than as definitive population parameters. To summarize,  $\tau_1$  through  $\tau_6$  are set to  $1.34\%$ ,  $0.5y$ ,  $y$ ,  $2y$ ,  $3y$ , and  $12.28\%$ , respectively, where  $y$  denotes the sample event rate ( $5.35\%$ ).

Table S1 presents the rare-events logistic regression estimates of the main effects of government trust on disaster volunteering across the six assumed values of  $\tau$ . Models S1 through S12 report the associations of each trust dimension entered separately, while Models S13 through S18 include both dimensions simultaneously. Each set is estimated across the six values of  $\tau$ , and the results are highly stable across these specifications throughout.

[Insert Table S1 Here]

The first set of models provides little evidence for the crowding-out expectation in H1. When entered alone, trust in national government is positively and significantly associated with disaster volunteering across all six specifications (Model S1,  $b = 0.362, p < 0.05$ ; Model A2,  $b = 0.362, p < 0.05$ ; Model S3,  $b = 0.363, p < 0.05$ ; Model S4,  $b = 0.363, p < 0.05$ ; Model S5,  $b = 0.363, p < 0.05$ ; Model S6,  $b = 0.363, p < 0.05$ ), and this association remains positive and significant after trust in local government is added (Model S13,  $b = 0.296, p < 0.10$ ; Model S14,  $b = 0.297, p < 0.10$ ; Model S15,  $b = 0.300, p < 0.10$ ; Model S16,  $b = 0.304, p < 0.10$ ; Model S17,  $b = 0.308, p < 0.10$ ; Model S18,  $b = 0.305, p < 0.10$ ). Rather than reducing citizens' likelihood of participation, higher trust in national government is associated with a greater likelihood of disaster volunteering. H1 is therefore not supported. Trust in local government also shows a positive association with disaster volunteering, though the relationship is partial. When entered separately, the coefficient is positive and significant across all six specifications (Model S7,  $b = 0.192, p < 0.05$ ; Model S8,  $b = 0.189, p < 0.05$ ; Model S9,  $b = 0.185, p < 0.05$ ; Model S10,  $b = 0.177, p < 0.05$ ; Model S11,  $b = 0.171, p < 0.05$ ; Model S12,  $b = 0.175, p < 0.05$ ). After trust in national government is included in the same model, the coefficient for local government trust becomes smaller and remains marginally significant for smaller assumed population event rates (Model S13,  $b = 0.140, p < 0.10$ ; Model S14,  $b = 0.137, p < 0.10$ ), while significance attenuates as  $\tau$  increases (Model S15,  $b = 0.132, p > 0.10$ ; Model S16,  $b = 0.123, p > 0.10$ ; Model S17,  $b = 0.116, p > 0.10$ ; Model S18,  $b = 0.121, p > 0.10$ ). In other words, H2 receives partial support. These patterns are consistent with the logit and Firth logit results reported in the main text.

Table S2 reports the rare-events logistic regression estimates of the context-moderated effects of government trust on disaster volunteering across the six values of  $\tau$  (Model S19,  $b = -0.640, p < 0.05$ ; Model S20,  $b = -0.641, p < 0.05$ ; Model S21,  $b = -0.643, p < 0.05$ ; Model S22,  $b = -0.648, p < 0.05$ ; Model S23,  $b = -0.653, p < 0.05$ ; Model S24,  $b = -0.649, p < 0.05$ ). The interaction between trust in national government and rural residence is negative and statistically significant across all six specifications, indicating that the positive association between trust in national government and disaster volunteering is attenuated in rural areas. The

interaction between trust in local government and rural residence is positive and statistically significant across all six specifications (Model S19,  $b = 0.418$ ,  $p < 0.05$ ; Model S20,  $b = 0.421$ ,  $p < 0.01$ ; Model S21,  $b = 0.426$ ,  $p < 0.01$ ; Model S22,  $b = 0.432$ ,  $p < 0.01$ ; Model S23,  $b = 0.437$ ,  $p < 0.01$ ; Model S24,  $b = 0.433$ ,  $p < 0.01$ ), indicating that the positive association between trust in local government and disaster volunteering is amplified in rural areas. These findings do not support H3a but provide support for H3b. Overall, the rare-events logit results are consistent with the main logit and Firth logit findings.

[Insert Table S2 Here]

**Table S1 Direct Effects of Trust in Government on Disaster Volunteering (Relogit)**

| Variables                        | Model                | Model                | Model                | Model                | Model                | Model                | Model                | Model                | Model                | Model                | Model                | Model                | Model                | Model                | Model                | Model                | Model                | Model                |
|----------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                  | S1                   | S2                   | S3                   | S4                   | S5                   | S6                   | S7                   | S8                   | S9                   | S10                  | S11                  | S12                  | S13                  | S14                  | S15                  | S16                  | S17                  | S18                  |
|                                  | τ1                   | τ2                   | τ3                   | τ4                   | τ5                   | τ6                   | τ1                   | τ2                   | τ3                   | τ4                   | τ5                   | τ6                   | τ1                   | τ2                   | τ3                   | τ4                   | τ5                   | τ6                   |
| Trust in national government     | 0.362**<br>(0.156)   | 0.362**<br>(0.156)   | 0.363**<br>(0.156)   | 0.363**<br>(0.155)   | 0.363**<br>(0.156)   | 0.363**<br>(0.155)   |                      |                      |                      |                      |                      |                      | 0.296*<br>(0.165)    | 0.297*<br>(0.165)    | 0.300*<br>(0.165)    | 0.304*<br>(0.164)    | 0.308*<br>(0.165)    | 0.305*<br>(0.164)    |
| Trust in local government        |                      |                      |                      |                      |                      |                      | 0.192**<br>(0.080)   | 0.189**<br>(0.079)   | 0.185**<br>(0.079)   | 0.177**<br>(0.078)   | 0.171**<br>(0.078)   | 0.175**<br>(0.078)   | 0.140*<br>(0.082)    | 0.137*<br>(0.082)    | 0.132<br>(0.081)     | 0.123<br>(0.080)     | 0.116<br>(0.081)     | 0.121<br>(0.080)     |
| Rural (0/1)                      | 0.104<br>(0.160)     | 0.103<br>(0.160)     | 0.102<br>(0.159)     | 0.099<br>(0.159)     | 0.096<br>(0.159)     | 0.098<br>(0.159)     | 0.093<br>(0.160)     | 0.091<br>(0.159)     | 0.087<br>(0.159)     | 0.078<br>(0.159)     | 0.071<br>(0.159)     | 0.076<br>(0.159)     | 0.095<br>(0.159)     | 0.094<br>(0.159)     | 0.091<br>(0.159)     | 0.085<br>(0.158)     | 0.081<br>(0.159)     | 0.084<br>(0.158)     |
| Male (0/1)                       | 1.139***<br>(0.155)  | 1.139***<br>(0.154)  | 1.138***<br>(0.154)  | 1.137***<br>(0.153)  | 1.135***<br>(0.153)  | 1.136***<br>(0.153)  | 1.193***<br>(0.156)  | 1.193***<br>(0.156)  | 1.191***<br>(0.155)  | 1.188***<br>(0.154)  | 1.185***<br>(0.154)  | 1.187***<br>(0.154)  | 1.164***<br>(0.157)  | 1.163***<br>(0.157)  | 1.161***<br>(0.156)  | 1.157***<br>(0.155)  | 1.153***<br>(0.155)  | 1.156***<br>(0.155)  |
| Age                              | -0.004<br>(0.006)    | -0.004<br>(0.006)    | -0.004<br>(0.006)    | -0.004<br>(0.006)    | -0.004<br>(0.006)    | -0.004<br>(0.006)    | -0.002<br>(0.006)    | -0.002<br>(0.006)    | -0.002<br>(0.006)    | -0.001<br>(0.006)    | -0.001<br>(0.006)    | -0.001<br>(0.006)    | -0.004<br>(0.006)    | -0.003<br>(0.006)    | -0.003<br>(0.006)    | -0.003<br>(0.006)    | -0.003<br>(0.006)    | -0.003<br>(0.006)    |
| Age <sup>2</sup>                 | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) |
| Education                        |                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |                      |
| Junior high school               | 0.096<br>(0.197)     | 0.098<br>(0.196)     | 0.103<br>(0.194)     | 0.112<br>(0.193)     | 0.120<br>(0.192)     | 0.114<br>(0.192)     | 0.108<br>(0.196)     | 0.110<br>(0.195)     | 0.114<br>(0.194)     | 0.121<br>(0.192)     | 0.129<br>(0.191)     | 0.124<br>(0.192)     | 0.105<br>(0.197)     | 0.107<br>(0.196)     | 0.112<br>(0.194)     | 0.120<br>(0.193)     | 0.127<br>(0.192)     | 0.122<br>(0.192)     |
| Senior high school or equivalent | 0.396*<br>(0.232)    | 0.400*<br>(0.231)    | 0.407*<br>(0.229)    | 0.418*<br>(0.226)    | 0.425*<br>(0.224)    | 0.420*<br>(0.225)    | 0.406*<br>(0.232)    | 0.410*<br>(0.230)    | 0.417*<br>(0.228)    | 0.428*<br>(0.225)    | 0.437*<br>(0.224)    | 0.431*<br>(0.225)    | 0.396*<br>(0.232)    | 0.400*<br>(0.231)    | 0.406*<br>(0.229)    | 0.417*<br>(0.226)    | 0.426*<br>(0.224)    | 0.420*<br>(0.225)    |
| Junior college                   | 0.332<br>(0.310)     | 0.339<br>(0.309)     | 0.351<br>(0.309)     | 0.370<br>(0.310)     | 0.385<br>(0.311)     | 0.374<br>(0.310)     | 0.335<br>(0.307)     | 0.340<br>(0.307)     | 0.349<br>(0.307)     | 0.366<br>(0.308)     | 0.380<br>(0.309)     | 0.370<br>(0.308)     | 0.315<br>(0.310)     | 0.322<br>(0.309)     | 0.335<br>(0.309)     | 0.355<br>(0.310)     | 0.372<br>(0.311)     | 0.361<br>(0.310)     |
| Bachelor's degree or above       | 0.075<br>(0.343)     | 0.078<br>(0.342)     | 0.086<br>(0.341)     | 0.099<br>(0.341)     | 0.111<br>(0.342)     | 0.103<br>(0.341)     | 0.095<br>(0.340)     | 0.097<br>(0.340)     | 0.101<br>(0.340)     | 0.112<br>(0.341)     | 0.124<br>(0.342)     | 0.116<br>(0.341)     | 0.072<br>(0.342)     | 0.074<br>(0.342)     | 0.078<br>(0.341)     | 0.088<br>(0.341)     | 0.097<br>(0.341)     | 0.090<br>(0.341)     |
| Married (0/1)                    | 0.094<br>(0.201)     | 0.090<br>(0.201)     | 0.083<br>(0.202)     | 0.071<br>(0.204)     | 0.062<br>(0.206)     | 0.068<br>(0.204)     | 0.095<br>(0.201)     | 0.091<br>(0.201)     | 0.084<br>(0.202)     | 0.073<br>(0.204)     | 0.062<br>(0.206)     | 0.069<br>(0.204)     | 0.100<br>(0.202)     | 0.095<br>(0.202)     | 0.087<br>(0.203)     | 0.074<br>(0.204)     | 0.064<br>(0.206)     | 0.071<br>(0.205)     |
|                                  | 0.099                | 0.098                | 0.097                | 0.094                | 0.092                | 0.093                | 0.124                | 0.123                | 0.120                | 0.117                | 0.114                | 0.116                | 0.109                | 0.107                | 0.105                | 0.102                | 0.099                | 0.101                |

|                                  |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |          |
|----------------------------------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Agricultural household (0/1)     | (0.180)  | (0.179)  | (0.178)  | (0.176)  | (0.175)  | (0.176)  | (0.179)  | (0.179)  | (0.178)  | (0.177)  | (0.176)  | (0.177)  | (0.180)  | (0.179)  | (0.178)  | (0.177)  | (0.176)  | (0.176)  |
| Communist Party membership (0/1) | 1.031*** | 1.029*** | 1.023*** | 1.015*** | 1.009*** | 1.013*** | 1.013*** | 1.011*** | 1.008*** | 1.003*** | 0.999*** | 1.002*** | 0.994*** | 0.992*** | 0.988*** | 0.983*** | 0.978*** | 0.981*** |
| Ethnic minority (0/1)            | -0.229   | -0.234   | -0.244   | -0.259   | -0.273   | -0.263   | -0.241   | -0.247   | -0.257   | -0.276   | -0.293   | -0.281   | -0.250   | -0.255   | -0.264   | -0.280   | -0.294   | -0.284   |
| GDP per capita (log)             | -0.072   | -0.079   | -0.092   | -0.112   | -0.127   | -0.117   | -0.028   | -0.037   | -0.053   | -0.080   | -0.101   | -0.087   | -0.039   | -0.048   | -0.062   | -0.086   | -0.104   | -0.092   |
| Disaster economic loss           | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  | 0.001**  |
| Constant                         | -6.259*  | -5.461   | -4.579   | -3.575   | -2.913   | -3.357   | -6.060*  | -5.234   | -4.295   | -3.184   | -2.427   | -2.937   | -6.826*  | -6.012*  | -5.098   | -4.038   | -3.329   | -3.806   |
| Observations                     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     | 4710     |
| Regional Control                 | YES      | YES      | YES      | YES      | YES      | YES      | YES      | YES      | YES      | YES      | YES      | YES      | YES      | YES      | YES      | YES      | YES      | YES      |

Notes: Standard errors are in parentheses. Age is mean-centered before being squared. Education is entered as dummy variables, with primary school or below as the reference category. Regional controls are included but not reported; North China is the reference category. \* p < 0.1, \*\* p < 0.05, \*\*\* p < 0.01.

Source: Authors

**Table S2 Moderating Effects of Trust in Government on Disaster Volunteering (Relogit)**

| Variables                        | Model S19            | Model S20            | Model S21            | Model S22            | Model S23            | Model S24            |
|----------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                  | $\tau 1$             | $\tau 2$             | $\tau 3$             | $\tau 4$             | $\tau 5$             | $\tau 6$             |
| Trust in national government     | 0.587***<br>(0.227)  | 0.589***<br>(0.227)  | 0.592***<br>(0.227)  | 0.599***<br>(0.228)  | 0.604***<br>(0.229)  | 0.600***<br>(0.228)  |
| Trust in local government        | -0.050<br>(0.107)    | -0.053<br>(0.106)    | -0.060<br>(0.105)    | -0.070<br>(0.104)    | -0.079<br>(0.105)    | -0.073<br>(0.104)    |
| Rural (0/1)                      | 0.134<br>(0.166)     | 0.133<br>(0.165)     | 0.132<br>(0.165)     | 0.128<br>(0.165)     | 0.123<br>(0.166)     | 0.126<br>(0.166)     |
| Trust in national government *   | -0.640**<br>(0.320)  | -0.641**<br>(0.320)  | -0.643**<br>(0.319)  | -0.648**<br>(0.319)  | -0.653**<br>(0.318)  | -0.649**<br>(0.319)  |
| Trust in local government *      | 0.418**<br>(0.163)   | 0.421***<br>(0.162)  | 0.426***<br>(0.161)  | 0.432***<br>(0.160)  | 0.437***<br>(0.160)  | 0.433***<br>(0.160)  |
| Male (0/1)                       | 1.163***<br>(0.157)  | 1.163***<br>(0.156)  | 1.162***<br>(0.155)  | 1.160***<br>(0.155)  | 1.158***<br>(0.154)  | 1.159***<br>(0.154)  |
| Age                              | -0.003<br>(0.006)    | -0.003<br>(0.006)    | -0.003<br>(0.006)    | -0.003<br>(0.006)    | -0.002<br>(0.006)    | -0.002<br>(0.006)    |
| Age <sup>2</sup>                 | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) | -0.002***<br>(0.000) |
| Education                        |                      |                      |                      |                      |                      |                      |
| Junior high school               | 0.103<br>(0.198)     | 0.104<br>(0.197)     | 0.106<br>(0.196)     | 0.112<br>(0.194)     | 0.118<br>(0.193)     | 0.114<br>(0.194)     |
| Senior high school or equivalent | 0.408*<br>(0.234)    | 0.409*<br>(0.232)    | 0.411*<br>(0.230)    | 0.413*<br>(0.226)    | 0.414*<br>(0.224)    | 0.413*<br>(0.225)    |
| Junior college                   | 0.332<br>(0.312)     | 0.339<br>(0.312)     | 0.350<br>(0.311)     | 0.367<br>(0.312)     | 0.380<br>(0.313)     | 0.371<br>(0.312)     |
| Bachelor's degree or above       | 0.108<br>(0.345)     | 0.109<br>(0.345)     | 0.112<br>(0.344)     | 0.118<br>(0.343)     | 0.125<br>(0.343)     | 0.120<br>(0.343)     |
| Married (0/1)                    | 0.088<br>(0.202)     | 0.083<br>(0.202)     | 0.075<br>(0.202)     | 0.064<br>(0.204)     | 0.055<br>(0.206)     | 0.061<br>(0.204)     |
| Agricultural household (0/1)     | 0.111<br>(0.179)     | 0.108<br>(0.179)     | 0.102<br>(0.178)     | 0.094<br>(0.176)     | 0.087<br>(0.176)     | 0.092<br>(0.176)     |
| Communist Party membership (0/1) | 0.991***<br>(0.195)  | 0.990***<br>(0.193)  | 0.989***<br>(0.191)  | 0.986***<br>(0.189)  | 0.983***<br>(0.188)  | 0.985***<br>(0.189)  |
| Ethnic minority (0/1)            | -0.241<br>(0.274)    | -0.247<br>(0.270)    | -0.258<br>(0.266)    | -0.277<br>(0.260)    | -0.294<br>(0.257)    | -0.282<br>(0.259)    |
| GDP per capita (log)             | -0.010<br>(0.309)    | -0.016<br>(0.309)    | -0.029<br>(0.308)    | -0.049<br>(0.309)    | -0.064<br>(0.311)    | -0.054<br>(0.310)    |
| Disaster economic loss           | 0.001**<br>(0.000)   | 0.001**<br>(0.000)   | 0.001**<br>(0.000)   | 0.001**<br>(0.000)   | 0.001**<br>(0.000)   | 0.001**<br>(0.000)   |
| Constant                         | -5.708*<br>(3.461)   | -4.912<br>(3.453)    | -4.027<br>(3.445)    | -3.012<br>(3.451)    | -2.339<br>(3.469)    | -2.791<br>(3.455)    |
| Observations                     | 4710                 | 4710                 | 4710                 | 4710                 | 4710                 | 4710                 |
| Regional Control                 | YES                  | YES                  | YES                  | YES                  | YES                  | YES                  |

Notes: Standard errors are in parentheses. Age is mean-centered before being squared. Education is entered as dummy variables, with primary school or below as the reference category. Regional controls are included but not reported; North China is the reference category. From Model S19 to A24, trust in national and local government variables are mean-centered before entering the interaction terms. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Source: Authors



## Supplemental Materials 2 Moderating Role of Nonprofit Sector

### Development

As a supplementary analysis, we examine whether provincial nonprofit sector development further conditions the relationship between trust in government and disaster volunteering within rural-urban split-samples. This analysis is intended to assess whether the rural-urban heterogeneity identified in the main analysis may be further clarified by differences in formal organizational pathways. Nonprofit sector development captures the extent to which the local institutional environment provides organized and formalized channels for civic and voluntary participation. A more developed nonprofit sector may increase the availability of organizational pathways through which willingness to participate can be translated into actual involvement (Simo and Bies 2007; Lecy and Van Slyke 2013). We measure nonprofit sector development as the per capita availability of nonprofit organizations at the provincial level (Matsunaga and Yamauchi 2004), calculated from the number of nonprofit organizations by province at the end of *Q4 2020*, obtained from the website of *Ministry of Civil Affairs of China* (2020), divided by provincial population from the *China Statistical Yearbook 2021* (National Bureau of Statistics of China 2021).

Because nonprofit sector development is measured at the provincial level, the main specification uses a multilevel logistic regression with random intercepts for provinces to account for the nesting of individuals within provinces and to allow the baseline probability of participation to vary across provinces (Zhu and Zhang 2019). As robustness checks, we also estimate a single-level logit model with province-clustered standard errors, which retains the baseline logit specification but adjusts the standard errors for within-province dependence (Liang and Zeger 1986), as well as Firth logit models.

Table S3 reports split-sample estimates of the moderating role of nonprofit-sector development in rural and urban contexts. Models A25 through A27 present the estimates for the rural subsample, while Models A28 through A30 present the estimates for the urban subsample. The results show clear rural-urban differences in the level of trust in government associated with disaster volunteering. In the rural subsample, trust in local government remains positively and significantly associated with disaster volunteering (Model S25,  $b = 0.382$ ,  $p <$

0.01; Model S26,  $b = 0.382, p < 0.01$ ; Model S27,  $b = 0.362, p < 0.01$ ), whereas trust in national government is not significant (Model S25,  $b = -0.012, p > 0.10$ ; Model S26,  $b = -0.012, p > 0.10$ ; Model S27,  $b = -0.039, p > 0.10$ ). Moreover, the interactions between nonprofit sector development and trust in national government (Model S25,  $b = 0.121, p > 0.10$ ; Model S26,  $b = 0.121, p > 0.10$ ; Model S27,  $b = 0.111, p > 0.10$ ) and trust in local government (Model S25,  $b = 0.031, p > 0.10$ ; Model S26,  $b = 0.031, p > 0.10$ ; Model S27,  $b = 0.023, p > 0.10$ ) are statistically insignificant. This suggests that, in rural settings, the conversion of trust in local government into disaster volunteering is less dependent on province-level nonprofit sector development, consistent with the argument that rural mobilization is more likely to operate through place-based social ties and informal relational channels.

In the urban subsample, by contrast, trust in national government is positively and significantly associated with disaster volunteering (Model S28,  $b = 0.656, p < 0.01$ ; Model S29,  $b = 0.660, p < 0.05$ ; Model S30,  $b = 0.629, p < 0.01$ ), whereas trust in local government is not significant as a main effect (Model S28,  $b = -0.054, p > 0.10$ ; Model S29,  $b = -0.053, p > 0.10$ ; Model S30,  $b = -0.059, p > 0.10$ ). The interaction between trust in local government and nonprofit sector development is positive across all three specifications and reaches conventional levels of statistical significance in the multilevel logit (Model S28,  $b = 0.115, p < 0.10$ ) and Firth logit models (Model S30,  $b = 0.111, p < 0.10$ ). This pattern provides qualified supplementary support for the interpretation that formal organizations may serve as institutional pathways through which trust in local government is converted into volunteering in urban areas.

[Insert Table S3 Here]

The split-sample analysis should be interpreted as supplementary. Provincial nonprofit density is a broad and distal measure of organizational context and may not fully capture disaster-specific volunteering capacity, citizens' direct exposure to recruitment opportunities, or the concrete channels through which emergency volunteering is activated. The results are consistent with the main argument that rural-urban heterogeneity reflects differences in the mobilization channels through which government trust becomes actionable. In rural areas, trust

in local government is more readily converted into volunteering through embedded social relations. In urban areas, formal organizational pathways may help compensate for weaker interpersonal ties and provide channels through which trust in local government can be translated into participation.

**Table S3 Moderating Effects of Nonprofit Sector Development on Government Trust and Disaster Volunteering in Rural and Urban Subsamples**

| Variables                        | Model S25            | Model S26                   | Model S27           | Model S28            | Model S29                   | Model S30            |
|----------------------------------|----------------------|-----------------------------|---------------------|----------------------|-----------------------------|----------------------|
|                                  | Rural                |                             |                     | Urban                |                             |                      |
|                                  | Multilevel<br>logit  | Logit with<br>clustered SEs | Firth logit         | Multilevel<br>logit  | Logit with<br>clustered SEs | Firth logit          |
| Trust in national government     | -0.012<br>(0.228)    | -0.012<br>(0.279)           | -0.039<br>(0.223)   | 0.656***<br>(0.221)  | 0.660**<br>(0.286)          | 0.629***<br>(0.217)  |
| Trust in local government        | 0.382***<br>(0.130)  | 0.382***<br>(0.141)         | 0.362***<br>(0.127) | -0.054<br>(0.109)    | -0.053<br>(0.117)           | -0.059<br>(0.107)    |
| Nonprofit sector development     | -0.146<br>(0.100)    | -0.146<br>(0.089)           | -0.127<br>(0.098)   | -0.016<br>(0.079)    | -0.013<br>(0.061)           | -0.004<br>(0.070)    |
| Trust in national government *   | 0.121<br>(0.153)     | 0.121<br>(0.154)            | 0.111<br>(0.153)    | -0.148<br>(0.103)    | -0.156<br>(0.114)           | -0.165<br>(0.102)    |
| Nonprofit sector development     | 0.031<br>(0.080)     | 0.031<br>(0.068)            | 0.023<br>(0.079)    | 0.115*<br>(0.059)    | 0.118<br>(0.074)            | 0.111*<br>(0.060)    |
| Trust in local government *      | 1.297***<br>(0.243)  | 1.297***<br>(0.286)         | 1.268***<br>(0.240) | 1.126***<br>(0.204)  | 1.117***<br>(0.266)         | 1.099***<br>(0.202)  |
| Male (0/1)                       | -0.006<br>(0.010)    | -0.006<br>(0.009)           | -0.006<br>(0.010)   | -0.001<br>(0.009)    | -0.001<br>(0.009)           | -0.000<br>(0.009)    |
| Age                              | -0.002***<br>(0.001) | -0.002**<br>(0.001)         | -0.002**<br>(0.001) | -0.002***<br>(0.001) | -0.002***<br>(0.001)        | -0.002***<br>(0.001) |
| Age <sup>2</sup>                 |                      |                             |                     |                      |                             |                      |
| Education                        |                      |                             |                     |                      |                             |                      |
| Junior high school               | -0.014<br>(0.255)    | -0.014<br>(0.302)           | -0.018<br>(0.251)   | 0.191<br>(0.317)     | 0.190<br>(0.344)            | 0.167<br>(0.311)     |
| Senior high school or equivalent | 0.444<br>(0.313)     | 0.444<br>(0.400)            | 0.437<br>(0.309)    | 0.379<br>(0.338)     | 0.377<br>(0.340)            | 0.351<br>(0.332)     |
| Junior college                   | 0.429<br>(0.509)     | 0.429<br>(0.687)            | 0.447<br>(0.496)    | 0.360<br>(0.404)     | 0.371<br>(0.410)            | 0.354<br>(0.396)     |
| Bachelor's degree or above       | 0.368<br>(0.651)     | 0.368<br>(0.863)            | 0.421<br>(0.622)    | 0.156<br>(0.424)     | 0.165<br>(0.396)            | 0.142<br>(0.417)     |
| Married (0/1)                    | 0.359<br>(0.352)     | 0.360<br>(0.389)            | 0.319<br>(0.345)    | -0.063<br>(0.266)    | -0.051<br>(0.219)           | -0.061<br>(0.261)    |
| Agricultural household (0/1)     | 0.114<br>(0.312)     | 0.114<br>(0.409)            | 0.089<br>(0.305)    | 0.080<br>(0.202)     | 0.076<br>(0.163)            | 0.073<br>(0.199)     |
| Communist Party membership (0/1) | 1.261***<br>(0.273)  | 1.261***<br>(0.420)         | 1.232***<br>(0.268) | 0.779***<br>(0.251)  | 0.780***<br>(0.244)         | 0.777***<br>(0.247)  |
| Ethnic minority (0/1)            | -0.231<br>(0.369)    | -0.231<br>(0.419)           | -0.176<br>(0.359)   | -0.098<br>(0.420)    | -0.107<br>(0.438)           | -0.046<br>(0.402)    |
| GDP per capita (log)             | 0.919<br>(0.574)     | 0.919<br>(0.630)            | 0.904<br>(0.563)    | -0.116<br>(0.482)    | -0.146<br>(0.453)           | -0.140<br>(0.441)    |
| Disaster economic loss           | 0.001*<br>(0.000)    | 0.001**<br>(0.000)          | 0.001*<br>(0.000)   | 0.001*<br>(0.000)    | 0.001*<br>(0.000)           | 0.001**<br>(0.000)   |

|                          |                      |                      |                      |                   |                   |                   |
|--------------------------|----------------------|----------------------|----------------------|-------------------|-------------------|-------------------|
| Constant                 | -14.560**<br>(6.345) | -14.560**<br>(7.030) | -14.190**<br>(6.220) | -3.211<br>(5.440) | -2.874<br>(5.059) | -2.787<br>(4.974) |
| Observations             | 2063                 | 2063                 | 2063                 | 2647              | 2647              | 2647              |
| Regional Control         | YES                  | YES                  | YES                  | YES               | YES               | YES               |
| Province/ Cluster        | 28                   | 28                   |                      | 30                | 30                |                   |
| Log likelihood           | -392.770             | -392.770             |                      | -485.571          | -485.810          |                   |
| Penalized log likelihood |                      |                      | -342.566             |                   |                   | -431.707          |
| Pseudo R <sup>2</sup>    |                      | 0.115                |                      |                   | 0.099             |                   |

Notes: Standard errors are in parentheses. Age is mean-centered before being squared. Education is entered as dummy variables, with primary school or below as the reference category. Regional controls are included but not reported; North China is the reference category. Standard errors are reported in parentheses. For the multilevel logit models, a random intercept is specified for province. For the single-level logit models, standard errors are clustered at the province level with small-sample correction. From Model S25 to A30, trust in national government, trust in local government and nonprofit sector development variables are mean-centered before entering the interaction terms. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Source: Authors

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